

Formulae and Manipulating Numbers

Variables: numbers represented by symbols

- Let's begin with something straightforward and familiar:
- Calculating speed, s
- This can be in:
- miles per hour (mph)
- metres per second (m/s)
- kilometres per hour (kph or km/h depending whom you ask!)
- anything else - just be consistent with your units.



Calculating speed

- s – speed, mph
- d – distance, miles
- t – time, hours
- $s = d/t$ (distance divided by time)
- So, if you travel 200 miles in 5 hours, your average speed is
- $200/5 = 40$



Rearranging variables

- We know $s = d/t$, but what if we want to calculate:
- distance for a known speed and time?
- You probably know already that we simply multiply speed by time to get distance, but let's look at the process:
- $s = d/t$ multiply both sides by t
- $st = td/t$
- $st = td/t$ the time variables on the righthand side cancel
- $st = d$, or
- $d = st$

Celsius and Fahrenheit

- To convert Celsius to Fahrenheit:
- Multiply the temperature in Celsius by 9
- Divide by 5
- Add 32

$$F = \frac{9C}{5} + 32$$



Rearrange to calculate Celsius

- Starting with the previous equation, we want C on its own.
- Subtract 32 from both sides.
- Multiply both sides by 5.
- Finally, divide both sides by 9.

• So

$$C = \frac{5}{9}(F - 32)$$

$$F = \frac{9C}{5} + 32$$

$$F - 32 = \frac{9C}{5} + 32 - 32 \rightarrow F - 32 = \frac{9C}{5}$$

$$5(F - 32) = 5\left(\frac{9C}{5}\right) \rightarrow 5(F - 32) = 9C$$

$$\frac{5}{9}(F - 32) = \frac{9}{9}C \rightarrow \frac{5}{9}(F - 32) = C$$

Don't forget BODMAS – order of precedence

- Sometimes called BIDMAS, PEDMAS or variants.
- B is for Brackets (P for Parentheses)
- I is for Index (E for Exponent), e.g. the 2 in x^2
- DM for Division, Multiplication (left to right precedence if both present)
- AS for Addition, Subtraction



BODMAS example

$$X = F \frac{(A + B)^C}{D - E} + G$$

This is a bit abstract, but we'll look at a concrete example next slide.

1. Add A and B.
2. Raise the result to the power C.
3. Subtract E from D. (Treat terms under a long line as if they were in brackets.)
4. Divide the result of Step 2 by the result of Step 3.
5. Multiply by F.
6. Add G.

BODMAS example

$$X = 3 \frac{(15 + 5)^2}{16 - 12} + 7$$

1. Add 15 and 5 to get 20.
2. Square it to get 400.
3. Subtract 12 from 16 to get 4.
4. Now divide 400 by 4 to get 100.
5. Multiply by 3 to get 300.
6. Finally, add 7 to get 307.

So, X is 307.

Compound interest

- Consider a bank that offers 5% interest on savings every year.
- For simplicity, assume that the interest is applied once per year.
- 5% can be expressed as 0.05.
- The formula for interest calculation is
 - $V = P(1 + R)^n$
- where
- V is value after n years
- P is the principal (initial) value
- R is interest rate
- n is the number of years.



$$V = P(1 + R)^n$$

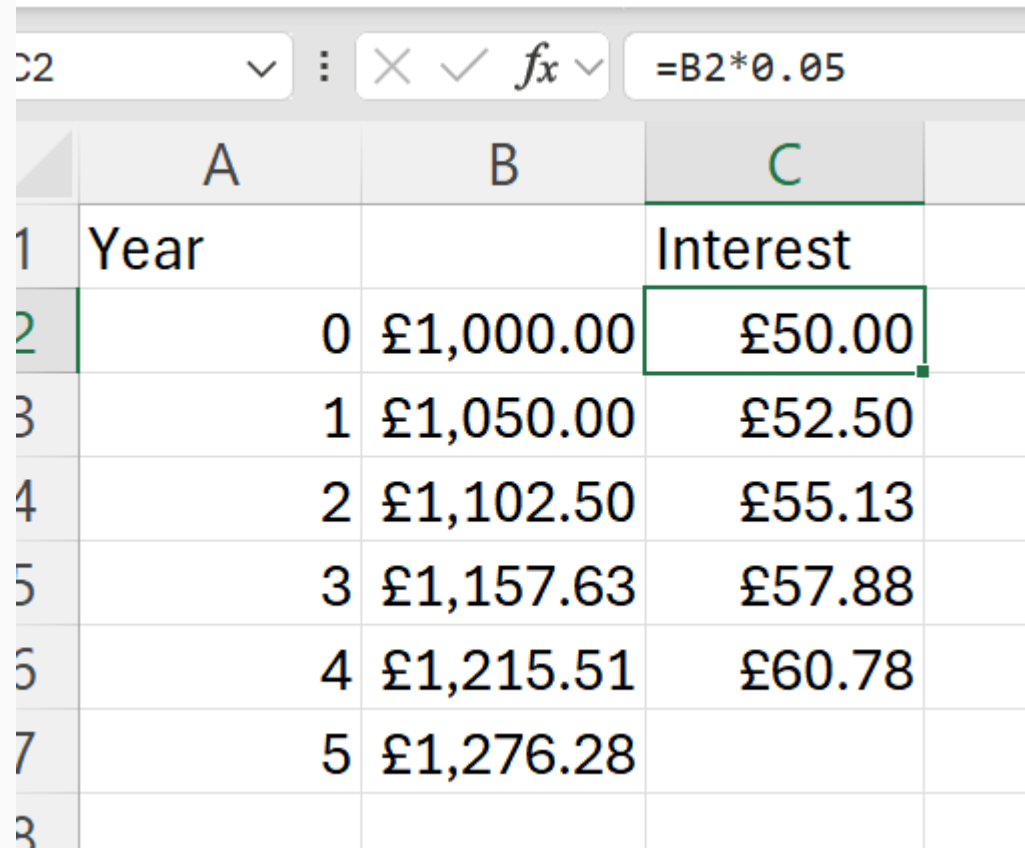
If you invest £1000, then after 5 years the value will be given as:

$$\begin{aligned} V &= 1000 * (1 + 0.05)^5 \\ &= 1000 * 1.05^5 \\ &= 1,276.28 \text{ to 2 decimal places} \end{aligned}$$

Note, the “to the power of” button on a scientific calculator looks like x^y , to the left of the 7:



Or you can use Excel



	A	B	C
1	Year		Interest
2	0	£1,000.00	£50.00
3	1	£1,050.00	£52.50
4	2	£1,102.50	£55.13
5	3	£1,157.63	£57.88
6	4	£1,215.51	£60.78
7	5	£1,276.28	
8			

Depreciation

- A similar formula is used for depreciation.
- For example, the value of a car that cost £8000 depreciates by 12% (0.12) per year.
- The formula is
 - $V = P(1 - D)^n$
- where P is the principal value and D is the depreciation rate.

Depreciation

- Using
 - $V = P(1 - D)^n$
- After 3 years, the value of the car is given as:
 - $V = 8000 * (1 - 0.12)^3$
- So, the car is worth £5,450 to the nearest £50.



Advanced manipulation of numbers

- You may wish to demonstrate that you can calculate the number of years required for a 5% investment to double your money.
- Using Excel is one way of doing this.
- Another makes use of the log key on a scientific calculator, near the bottom left.
- Note, you can use log or ln, but you must be consistent.
- A feature of logs is that $\log(X)^n$ is the same as $n \cdot \log(X)$

Years of interest

- Using the now familiar formula $V = P(1 + R)^n$
- If the value V is double P , then $V = 2P$.
- $2P = P(1 + R)^n$
- Divide both sides by P :
- $2 = (1 + R)^n$
- Taking logs of both sides,
- $\log(2) = \log(1 + R)^n$
- Using the aforementioned feature,
- $\log(2) = n \cdot \log(1 + R)$



Finding n

- What can we do with
- $\log(2) = n * \log(1 + R)$?
- Let's swap the terms:
- $n * \log(1 + R) = \log(2)$
- Divide both sides by $\log(1 + R)$
- The $\log(1 + R)$ expressions cancel

$$n \frac{\log(1 + R)}{\log(1 + R)} = \frac{\log(2)}{\log(1 + R)}$$

$$n = \frac{\log(2)}{\log(1 + R)}$$

Calculating n

$$n = \frac{\log(2)}{\log(1 + R)}$$

- Now it's a simple matter of entering the values:

$$n = \frac{\log(2)}{\log(1 + 0.05)}$$

- This gives an answer of 14.21 to 2 decimal places.
- So, we must wait 15 years for the value to double.